

Artificial Intelligence II
Deep Learning for Natural Language Processing
Spring Semester 2025-2026
Homework 4
Weight: 25% of Course Grade
Announced: May 22, 2026
Due: Jun 9, 2026, before 23:55

Description

You will develop a classifier for the task of **response clarity classification** using an agentic approach with instruction-tuned language models, rather than gradient-based fine-tuning. The dataset consists of political question-answer pairs (CLARITY dataset). Each instance contains a question and its corresponding answer, and your goal is to classify the answer into one of the following categories: *Clear Reply*, *Ambivalent*, or *Clear Non-Reply*.

The main focus of this homework is to design and evaluate a lightweight multi-agent framework for response clarity classification. The required method is called **D3-Agentic Prompting** and should be implemented with a small instruction-tuned model, specifically Qwen/Qwen3.5-0.8B. The goal is not to rely on repeated voting, but to test whether decomposing the classification problem into distinct agentic subtasks improves performance and interpretability.

Your D3-Agentic Prompting system must be compared against the methods you developed in the previous assignments. These previous systems should serve as the baselines for this homework, including vector-based classifiers and encoder-only transformer models. You may additionally experiment with other language models or agent variants if you wish, but these are optional extensions and not the main requirement.

Methodology: D3-Agentic Prompting

D3-Agentic Prompting is a lightweight multi-agent prompting framework in which several specialized agents perform different subtasks before a final classification decision is made. It should not be implemented as repeated voting over the same prompt. Instead, each agent should receive a targeted instruction and produce an intermediate output that supports the next stage of the pipeline.

The framework must include the following four agents:

1. **Question Intent Agent:** Identifies what the question asks for and what information would be required for a clear answer.
2. **Answer Content Agent:** Extracts what the answer actually says, focusing on explicit claims, omissions, and relevant details.
3. **Gap and Evasion Agent:** Compares the expected answer content with the actual answer content and detects missing elements, partial answers, ambiguity, or evasive patterns.

4. **Decision Agent:** Assigns the final label: *Clear Reply*, *Ambivalent*, or *Clear Non-Reply*, using the outputs of the previous agents.

Your experiments should test whether this decomposition of response clarity improves classification compared to the systems developed in the previous assignments. Concretely, implement D3-Agentic Prompting with Qwen/Qwen3.5-0.8B, run it on the same task, and compare its results with your previous vector-based and encoder-only models. You may optionally expand the architecture with additional agents, for example agents that retrieve or identify missing background information about persons, organizations, or events. You may also try additional language models beyond Qwen/Qwen3.5-0.8B. These extensions should be treated as optional ways to improve results, not as the main requirement. The primary objective is to implement and analyze the four-agent D3-Agentic Prompting architecture described above.

You should use the Hugging Face Transformers ecosystem and any supporting Python tools that help you run and evaluate prompted inference pipelines. The emphasis is on careful experimental design, prompt construction, and analysis of model behavior.

Your goal is to develop a strong D3-Agentic Prompting system, but also to understand the behavior of the individual agents and their contribution to the final decision. It is your responsibility to decide how to structure each agent prompt, how to pass intermediate outputs between agents, how to constrain outputs to valid labels, how to parse model generations robustly, and how to ensure that your conclusions are well supported by evidence.

Additionally, conduct a systematic comparison with the results obtained from the previous assessments (i.e., vector-based classifiers, encoder-only models, single LLM approaches). Analyze the differences in overall performance as well as in the observed failure modes. To what extent do the methods exhibit similar error patterns? Do they succeed and fail on the same instances, or do they demonstrate complementary behaviors across different cases? Your analysis should explicitly relate these findings to the earlier results, highlighting whether the prompting-based approach shares common limitations with prior methods or reveals distinct strengths and weaknesses.

Guidelines

This task consists of the following main steps:

1. **Agent design:** Decide how you will present the question-answer pair to Qwen/Qwen3.5-0.8B and how each D3 agent will receive and produce information. State your agent prompts explicitly.
2. **Previous-assignment baselines:** Use the systems you developed in the previous assignments as baselines, including vector-based classifiers and encoder-only transformer models.
3. **D3-Agentic Prompting:** Implement the four-agent architecture using Qwen/Qwen3.5-0.8B and clearly document the prompt, input, output format, and role of each agent.
4. **Inference pipeline:** Decide how you will handle generation settings, output constraints, invalid outputs, post-processing, and reproducibility.
5. **Evaluation and comparison:** Compare D3-Agentic Prompting against the previous-assignment baselines and justify your design choices clearly. On Kaggle, the primary evaluation metric is **F1-score**, while **Macro F1-score** should be reported as an additional metric. In your report, also include **accuracy**, **precision**, **recall**, and **F1-score**.
6. **Error analysis:** Analyze failure cases and discuss which agents, intermediate decisions, or prompting choices make some examples easier or harder.

It is your responsibility to choose all details of developing an effective prompting-based system, including prompt wording, demonstration selection, decoding decisions, output normalization, validation setup, and the criteria used to compare methods fairly.

For implementation purposes, students **must use a random seed** to ensure reproducibility.

Presentation of Results: Ensure that your results are supported with clear, high-quality, and well-labeled plots or tables that effectively illustrate your findings. For example, compare previous-assignment baselines, D3 agent outputs, decoding choices, optional agent extensions, and performance across different subsets of the data.

Additional Considerations: You are encouraged to mention in the appendices of your report any other prompting or inference approaches you explored that did not improve performance but contributed to your understanding of the task.

Experiments

You must conduct a systematic comparison between D3-Agentic Prompting and the methods developed in the previous assignments. Your experimental analysis should go beyond reporting final scores. In particular, you should investigate:

- the effect of prompt formulation choices for each D3 agent,
- the effect of decomposing the task into specialized agents,
- the quality and usefulness of each agent’s intermediate output,
- how D3-Agentic Prompting compares to vector-based classifiers and encoder-only transformer models,
- whether optional agent extensions or additional language models improve results,
- performance across different data subgroups (e.g., categories based on question length and answer length).

Your discussion must explain observed differences in performance rather than simply listing numerical results.

Error Analysis

A central component of this assignment is the analysis of model failures. You must perform a detailed error analysis and discuss questions such as:

- Which class is hardest to predict?
- Are there recurring linguistic patterns in incorrect predictions?
- Does D3-Agentic Prompting fail differently from the previous-assignment baselines?
- Do the different agents introduce or reduce specific types of errors?
- What is the fundamental source of error: what do prompted models fail to understand in question–answer interactions?
- What characteristics would an ideal prompting or reasoning system need in order to improve performance on this task (even if not implemented in this homework)?

You must include concrete examples of errors and organize your discussion in a meaningful way. In addition, you must describe:

- what changes you attempted in order to improve your agents, prompts, or inference pipeline,
- whether these changes improved performance,

- why you believe these changes helped or failed.

The main objective is not only to obtain a strong classifier, but also to understand why previous-assignment methods and D3-Agentic Prompting succeed or fail on this task.

Kaggle Competition

Submit your code as a **Jupyter Notebook** via the Kaggle competition. Follow these rules:

- Your team name must be your academic identification number (Αριθμός Μητρώου - sdiXXXXYYY or the one for graduate students).
- You must create and submit **one Kaggle notebook** that contains your full analysis, comparisons, and final submission pipeline.
- The notebook must include a systematic comparison of the previous-assignment baselines and D3-Agentic Prompting, and it must clearly identify the **best-performing approach**.
- Notebook naming format (use your academic ID): `[academic-id] homework-4-d3-agentic`. Share the notebook on Kaggle with the Teaching Assistant responsible for grading this assignment (Kaggle username: `konstantinosplas`).
- Your notebook must output a prediction file in csv format named, for example, `submission_best_d3-agentic_system.csv`. Do **not** submit as a file upload.
- On Kaggle, the primary metric is **F1-score**; report **Macro F1-score** as an additional metric in your analysis.
- The resulting file must follow the format specified in the provided `sample_submission.csv` file and must contain the predictions of your **best D3-Agentic or baseline system** over the test set.
- In your report, clearly state which approach performed best overall and explain in detail why, based on your evaluation findings.
- **Do not share your notebook publicly.**

Report

For this project, you are asked to create a detailed report. For this reason, we provide you with a L^AT_EX template. You may use the Overleaf online editor. Open Overleaf, create an account if you do not already have one, and upload the provided zip file by selecting *New Project* and then *Upload Project* (the template uses the pdfLaTeX compiler). If you encounter difficulties writing in L^AT_EX, you may initially draft your report in another editor following the structure of the template; however, you are strongly encouraged to produce the final version in L^AT_EX, as tools such as Overleaf or Prism can significantly simplify the writing process.

Your report must include: a description of the model and prompts you used; your D3-Agentic Prompting design choices; the previous-assignment baselines you used; your evaluation results; a comparative analysis of these baselines and D3-Agentic Prompting; a detailed error analysis; and a discussion of the modifications you tried and their effect.

The emphasis of the report is not only on predictive performance, but also on: understanding model behavior under prompting; explaining performance differences between the previous-assignment methods and D3-Agentic Prompting; analyzing model and agentic-pipeline limitations; and justifying experimental decisions.

The report must be written in English for students in the Master's program in Data Science and Information Technologies. All other students may choose their preferred language. The report must include the link to your Kaggle notebook, which must be shared with the teaching assistant and must not be made publicly available.

Grading

Implementation: Code, Kaggle submission [**Total 70%**]

- Data handling and preprocessing: [10%]
- Prompting pipeline development: [20%]
- Experiments: [30%]
- Prompting & inference optimization: [10%]

Report: Analysis and Presentation [**Total 30%**]

- Experiments: [10%]
- Analysis: [15%]
- Plots: [5%]

Important grading note: Code submissions without clear previous-baseline-vs-D3-Agentic analysis in the report will **not** receive implementation credit.

Submission Guides

We expect you to:

1. Submit your **executed Jupyter Notebook** (with outputs) and make it available to supervisors in **Kaggle** only.
2. Submit your report in **.pdf** format in e-class. Name your report as **[full-id].pdf**.

For example: ZZZZZZXXXXYYY.pdf

**We won't accept code submissions from e-class, e-mails, etc.*

Support

Konstantinos Plas is supervising this assignment. For any questions, please post them on Piazza.